

Intro to Machine Learning

Part of **Data Science Starter** · Estimated time: 55 min

Introduction

Machine learning, at its simplest, is teaching a program to find a pattern in examples so it can make predictions on new, unseen data.

Core ideas

- Features are the inputs a model learns from (e.g. square footage, number of bedrooms); the label is what it's trying to predict (e.g. house price).
- A train/test split holds back some data the model never sees during training, so you can honestly check whether it generalizes rather than just memorizing.
- scikit-learn's pattern is consistent across models: `model.fit(X_train, y_train)` to train, `model.predict(X_test)` to get predictions.
- Accuracy alone can be misleading — a model that always predicts "no" can score 95% accuracy on data where only 5% of cases are "yes", while being useless.

Example

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
model = LinearRegression().fit(X_train, y_train)
print(model.score(X_test, y_test))
```

Try it yourself

Using scikit-learn's built-in sample datasets (e.g. `sklearn.datasets.load_diabetes`), split the data, train a simple regression model, and print its score on the held-out test set.

Further reading

- [scikit-learn — Getting Started](https://scikit-learn.org/stable/getting_started.html)

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- [Google — Machine Learning Crash Course](https://developers.google.com/machine-learning/crash-course)

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